

Analog Input MAPS Template (Instrumentation)

The Analog Input MAPS template is designed to control an analog that is connected to an analog control unit.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in three models – advanced, standard and basic, which provide respectively lower levels of control over the analog input.

This document describes the Advanced Analog Input MAPS Template and also defines the level of functionality provided by the standard and basic models. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the analog being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Note: The version 2 (v2) Analog Input MAPS template, communicates Floating point values instead of Integer values to the driver. The High-High and Low-Low Alarms does not latch as does not set the FAULT active. The Fault Active is only when the signal is out of range, meaning over or under range.

Advanced Analog Input Functional Philosophy

The analog input is scaled and alarmed according to the setpoints from the Setup faceplate.

Field inputs

- Analog Input

PLC Setpoints

- High-high Alarm
- High Alarm
- Low Alarm
- Low-low Alarm

Simulation mode - will override input to the Analog Input and simulate an analog in the PLC program that is controlled from the faceplate. Simulation mode is used to test the process logic without receiving the physical input itself. It is very useful in testing, commissioning and maintenance.

Typical Faceplate Graphics for the Advanced Analog Input MAPS Template

Analog Input					
					
0000_ABCDE_123		Tag name			
123456		Analog indication value			
		Fault active			
		Simulation mode			
		High-High and Low-Low Alarm Active (Version 2 only)			
		High and Low Alarm Active (Version 2 only)			
Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshal	x	x	x
SSCL 1	High-high Alarm	Analog	x	x	
SSCL 2	High Alarm	Analog	x	x	
SSCL 3	Low Alarm	Analog	x	x	
SSCL 4	Low-low Alarm	Analog	x	x	
SSCL 5	Scale Min In Setpoint (Raw Min)	Analog	x		
SSCL 6	Scale Max In Setpoint (Raw Max)	Analog	x		
SSCL 7	Scale Min Out Setpoint (Eng. Min)	Analog	x		
SSCL 8	Scale Max Out Setpoint (Eng. Max)	Analog	x		
SSCL 9	Simulation Value	Analog	x	x	x
SSCL 10	Alarm Hysteresis	Analog	x	x	

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshal	x	x	x
SSSL 1	Analog Input Process Value (Scaled)	Analog	x	x	x
SSSL 2	Analog Input RAW Value	Analog	x		

Analog Input:				
AI 0	Analog Input for Analog Input	x	x	x

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Simulation	x	x	x	x
Bit 1	High-high AI Alarm Enabled	x	x	x	
Bit 2	High AI Alarm Enabled	x	x	x	
Bit 3	Low AI Alarm Enabled	x	x	x	
Bit 4	Low-Low AI Alarm Enabled	x	x	x	
Bit 5	Reset Fault		x	x	x

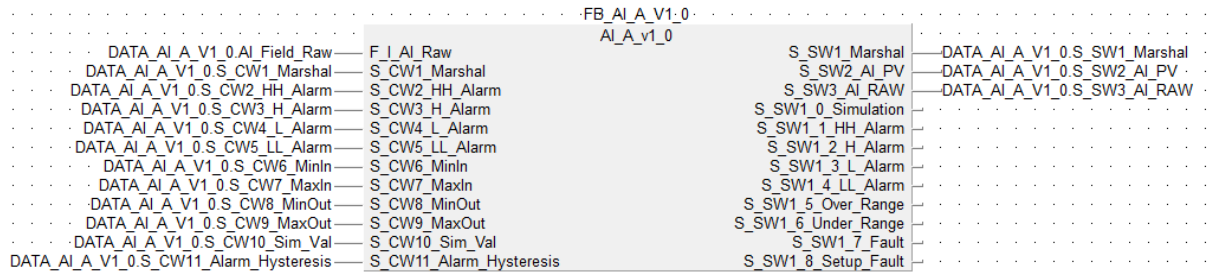
Bit 6	Spare				
Bit 7	Spare				
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Simulation On		x	x	x
Bit 1	High-high Alarm Active	x	x	x	
Bit 2	High Alarm Active	x	x	x	
Bit 3	Low Alarm Active	x	x	x	
Bit 4	Low-low Alarm Active	x	x	x	
Bit 5	Warning Active	x	x	x	
Bit 6	Alarm Active	x	x	x	
Bit 7	Alarm Fault Latch		x	x	x
Bit 8	Alarm Setup Fault		x	x	
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

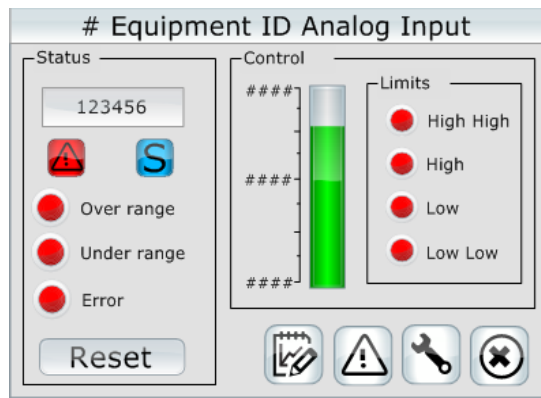
Resources	Advanced	Standard	Basic
Digital Inputs	0	0	0
Digital Outputs	0	0	0
Analog Inputs	1	1	1
Analog Outputs	0	0	0
PLC Memory Steps	359	173	86
System Words Used	40	3	2
System Bits Used	72	61	56
SCADA SCAN Points	14	9	4

Template: Advanced Analog Input PLC Function Block

<Device> (Advanced Analog Input Control)

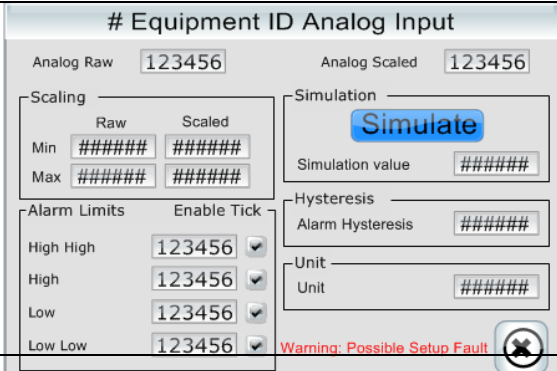


Template Graphic: Advanced Home Screen



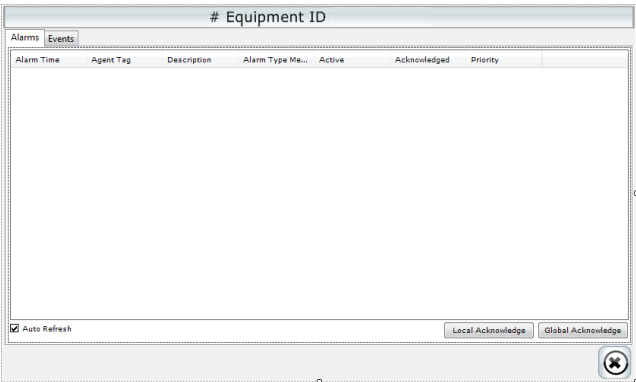
Control	Descriptions
# Equipment ID	Analog Input controlled name
High High	High-high alarm indication
High	High alarm indication
Low	Low alarm indication
Low Low	Low-low alarm indication
Over range	Analog over range alarm
Under range	Analog under range alarm
Error	Alarm fault latch
Reset	Reset the analog Input after a fault has been repaired
123456	Analog scaled value indication
S	Simulation indication
Warning	Fault latch

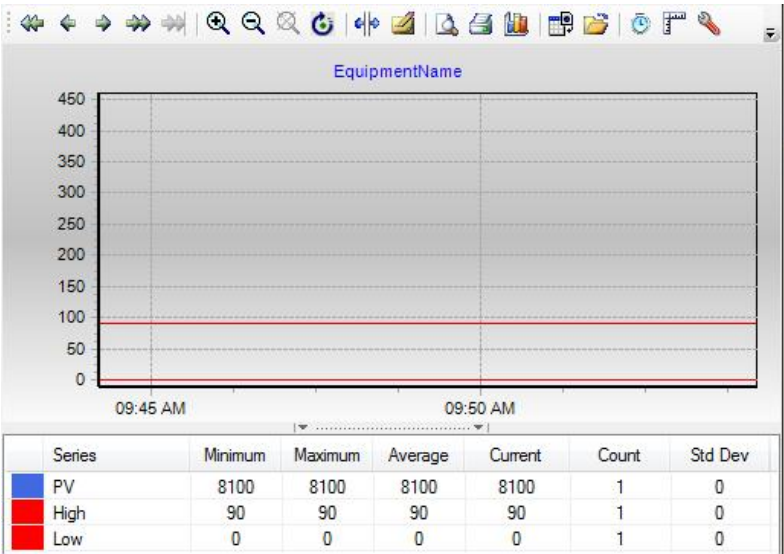
Template Graphic: Advanced Setup Screen



Control	Descriptions
Simulate	Enable simulation of analog Input, disables the physical output
Simulation Value	Simulation value of analog Input
Analog Raw	Raw value from field level
Analog Scaled	Scaled value from field level
Min, Max Raw	Raw level input values for scaling
Min, Max Scaled	Engineering level values for scaling
High High	High-high alarm setup value and enabling
High	High warning setup value and enabling
Low	Low warning setup value and enabling

	Low Low	Low-low alarm setup value and enabling
	Alarm Hysteresis	High and low warning auto reset hysteresis value. This is a value at which the warnings will reset should the process value decrease – in the case of the high level or increase - in the case of a low level
	Unit	Unit displayed for the scaled analog

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Equipment Name	Display the name of the current analog Input of the trend
	PV	Display the level of the current level
	High	High Setpoint for analog scaled value
	Low	Low Setpoint for analog scaled value